

POSM Polarization Tap and Stokes Polarimeter

Onboarding Packet

v1 Draft — Modular Power Meter, Polarization Monitor, and Full-Stokes Diagnostic Node

Photonics and Optical Sciences Club at Minnesota (POSM)

Draft date: June 11, 2026

Project mission. Build a compact optical diagnostic module that samples a small fraction of a beam, analyzes its polarization with a motorized quarter-wave plate and polarizing beamsplitter, detects the two output powers with photodiodes, digitizes both channels, and streams timestamped data to the FPGA/base-control system. The final instrument should operate as a power meter, a polarization-projection monitor, and a full-Stokes polarimeter for automated polarization diagnostics and feedback.

Baseline POSM use case. The first deployment is in the rubidium MTS probe path, immediately before the spectroscopy cell. The module directly measures the probe beam entering the cell. Pump-path inference is possible only through the known optical model of the system.

How to read this packet. Sections 1–5 explain the project motivation, polarization basics, and optical signal path. Sections 6–10 explain the photodiode, transimpedance amplifier, ADC, connector, and FPGA/data roles. Sections 11–14 describe calibration, mechanical design, acceptance criteria, and useful resources.

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1 Start Here: What Are We Building?

This packet assumes no prior background in polarization optics, photodiode circuits, polarimetry, or laboratory diagnostic modules. The goal is to make the project understandable from the ground up.

1.1 The one-sentence version

We are building a small module that steals a small fraction of an optical beam, measures how much light is present, measures how that light is polarized, and reports the result digitally so POSM experiments can monitor and eventually correct optical drift.

1.2 Why this matters

Optical experiments are fragile. A beam can have the correct color and path while still being wrong because:

- the total optical power drifts,
- the polarization rotates because of waveplates, fiber stress, mirrors, or thermal drift,
- a beamsplitter or PBS sends the wrong amount of light into a path,
- the experiment appears broken even though the laser is fine, because the polarization entering the cell is wrong.

The polarization tap is meant to turn those hidden optical variables into measured data. Instead of guessing whether the beam entering the spectroscopy cell has the correct power and polarization, the experiment can read two ADC channels and reconstruct the relevant optical state.

POSM note

The MTS setup is a good first application because power and polarization affect signal size, background, lock quality, and long-term reproducibility. The same module should later be usable anywhere POSM wants a compact power/polarization diagnostic.

1.3 What the module does and does not do

Does	Does not
Samples a small part of the beam	Replace the main beam path
Measures total sampled power	Lock the laser by itself
Measures a polarization projection continuously	Drive the QWP motor directly; that is the separate piezomotor project
Supports full-Stokes snapshots by stepping the QWP	Require high-speed RF photodetection
Streams timestamped ADC data to the FPGA/base system	Require FPGA-side Stokes fitting in v1
Can support automated polarization feedback later	Need continuous QWP spinning all day

2 Polarization Basics for This Project

2.1 What is polarization?

Light is an electromagnetic wave. For a beam traveling along one direction, the electric field oscillates in the transverse plane. The path traced by the electric-field vector determines the polarization:

- **Linear polarization:** the electric field oscillates along a fixed axis.
- **Circular polarization:** the electric field rotates with constant magnitude.
- **Elliptical polarization:** the general case; the electric field traces an ellipse.

In the lab, polarization matters because many optical components are polarization-dependent. A PBS transmits one linear polarization and reflects the orthogonal one. A waveplate changes the relative phase between polarization components. Atomic spectroscopy signals can change if the beam polarization changes.

2.2 Stokes parameters

A convenient way to quantify polarization is with the Stokes vector:

$$\mathbf{S} = \begin{bmatrix} S_0 \\ S_1 \\ S_2 \\ S_3 \end{bmatrix}. \quad (1)$$

A useful interpretation is:

Parameter	Meaning
S_0	Total optical intensity or power.
S_1	Horizontal vs. vertical linear polarization balance.
S_2	$+45^\circ$ vs. -45° linear polarization balance.
S_3	Right-circular vs. left-circular polarization balance.

The normalized Stokes parameters are

$$s_i = \frac{S_i}{S_0}, \quad i \in \{1, 2, 3\}. \quad (2)$$

The normalized parameters describe polarization independent of total power, which is useful because beam power can drift even when polarization is stable.

2.3 What a PBS measurement gives you

A polarizing beamsplitter (PBS) separates the incoming beam into two orthogonal linear polarization components. If the two photodiodes measure powers P_1 and P_2 , then

$$P_{\text{sum}} = P_1 + P_2 \quad (3)$$

is proportional to total sampled power, while

$$D = \frac{P_1 - P_2}{P_1 + P_2} \quad (4)$$

is a normalized polarization projection. A fixed PBS alone gives one projection of the polarization state, not the full Stokes vector.

2.4 Why we add a quarter-wave plate

The quarter-wave plate (QWP) changes the polarization before the PBS. By rotating the QWP to several known angles and measuring the PBS output powers at each angle, the instrument can fit for the full Stokes vector.

For an ideal rotating QWP followed by a linear analyzer, the normalized detector difference has sinusoidal components such as

$$d(\theta) = \frac{I_1(\theta) - I_2(\theta)}{I_1(\theta) + I_2(\theta)} \approx a_0 + a_1 \cos(4\theta) + a_2 \sin(4\theta) + a_3 \sin(2\theta), \quad (5)$$

where θ is the QWP angle. The fitted coefficients map to the Stokes parameters after calibration. The exact signs depend on the QWP angle convention, PBS port labeling, and photodiode wiring, so calibration is required.

Important

Continuous high-duty-cycle QWP dithering is not required for v1. The preferred operating mode is fixed-angle monitoring plus occasional stepped-Stokes measurements. This protects the piezomotor and keeps the measurement easy to interpret.

3 System Overview

3.1 Optical and electrical chain

The full chain is:

```
Probe beam before cell
-> BP058 beam sampler
-> motorized QWP, WPQ05M-780
-> 12.7 mm PBS cube, Shalom EO 20707-002
-> FDS1010 photodiode A and B
-> OPA2381 dual transimpedance amplifier
-> ADS8353 dual simultaneous ADC
-> DF40 mezzanine connection to base PCB
-> FPGA SPI readout and timestamping
-> host/POSMq Stokes reconstruction and feedback logic
```

3.2 Baseline use in the MTS setup

The module will initially sit in the probe path before the spectroscopy cell. The MTS system may have pump and probe beams, but this tap directly samples the probe beam only. Pump-path information is inferred only if the pump/probe relationship is known from the optical model.

3.3 Power budget

The frozen design assumption is:

$$P_{\text{MTS, total}} \leq 5 \text{ mW}. \quad (6)$$

Assuming the pump/probe split is approximately 50/50, the maximum probe power entering the tap is

$$P_{\text{probe,max}} \approx 2.5 \text{ mW}. \quad (7)$$

The BP058 sampler is treated as an 8 percent sampler:

$$P_{\text{sample}} = 0.08 P_{\text{probe}}. \quad (8)$$

Therefore,

$$P_{\text{sample,max}} = 0.08 \times 2.5 \text{ mW} = 0.2 \text{ mW} = 200 \mu\text{W}. \quad (9)$$

Worst case for one photodiode occurs when the PBS sends nearly all sampled power to one output port:

$$P_{\text{PD,max}} \approx 200 \mu\text{W}. \quad (10)$$

3.4 Power and transimpedance design table

For a rough design calculation, use silicon photodiode responsivity near 780 nm of approximately

$$R_\lambda \approx 0.5 \text{ A/W}. \quad (11)$$

Then

$$I_{\text{PD}} = R_\lambda P_{\text{PD}}. \quad (12)$$

$P_{\text{MTS,total}}$	P_{probe}	P_{sample}	$P_{\text{PD,worst}}$	I_{PD}	Z_T for 1 V	Z_T for 2 V
0.5 mW	0.25 mW	20 μW	20 μW	10 μA	100 k Ω	200 k Ω
1.0 mW	0.50 mW	40 μW	40 μW	20 μA	50 k Ω	100 k Ω
2.0 mW	1.00 mW	80 μW	80 μW	40 μA	25 k Ω	50 k Ω
3.0 mW	1.50 mW	120 μW	120 μW	60 μA	16.7 k Ω	33.3 k Ω
5.0 mW	2.50 mW	200 μW	200 μW	100 μA	10 k Ω	20 k Ω

POSM note

The student should use this table as the starting point for the feedback resistor calculation. A transimpedance near 20 k Ω gives roughly 2 V output at the frozen worst-case one-photodiode power. The exact R_f and C_f values should be calculated and verified during schematic design.

4 Optical Subsystem

4.1 Beam sampler: Thorlabs BP058

The BP058 samples a small fraction of the probe beam so the diagnostic module does not significantly disturb the main beam entering the cell. In this design, the sampler is treated as an 8:92 reflected/transmitted pellicle sampler over the relevant wavelength range.

4.2 Quarter-wave plate: Thorlabs WPQ05M-780

The QWP is the polarization-analysis element. At a fixed angle, it lets the PBS measure one polarization projection. By stepping the QWP through multiple angles, the system can estimate the full Stokes vector. The QWP is mechanically driven by the separate piezomotor rotation project; the polarization-tap electronics only record the optical response and timestamp it.

4.3 Polarizing beamsplitter: Shalom EO 20707-002

The PBS separates the analyzed beam into two orthogonal polarization components. Each output goes to one FDS1010 photodiode. The part chosen for the current design is Shalom EO 20707-002, a 12.7 mm PBS cube for the 650–950 nm band.

4.4 Photodiode placement

The FDS1010 active area is large, so the mechanical alignment is forgiving. However, the photodiodes should still be held in repeatable pockets or slots. Do not depend on glue as the final alignment method.

The desired layout is:

```
BP058 sampler -> QWP -> PBS
                  | -> FDS1010 A
                  | -> FDS1010 B
```

5 Photodiode Basics and FDS1010 Use

5.1 What a photodiode does

A photodiode converts optical power into photocurrent. For a given wavelength,

$$I_{PD} = R_{\lambda}P, \quad (13)$$

where R_{λ} is responsivity in A/W and P is optical power. If $R_{\lambda} = 0.5$ A/W and $P = 100$ μ W, then

$$I_{PD} = 0.5 \text{ A/W} \times 100 \text{ } \mu\text{W} = 50 \text{ } \mu\text{A}. \quad (14)$$

5.2 FDS1010 design choice

The FDS1010 is used because it has a large active area and is easy to align to the PBS output beams. The selected operating mode for POSM is:

$$\text{FDS1010 reverse bias} = 5 \text{ V}. \quad (15)$$

The reverse bias reduces junction capacitance compared with zero-bias operation and matches common FDS1010 datasheet conditions. The expected signal bandwidth is only tens of kHz, so the photodiode speed is not the limiting factor.

5.3 Photodiode terminal convention

For the selected photoconductive TIA topology:

```
FDS1010 cathode -> +5 V reverse-bias network
FDS1010 anode   -> OPA2381 inverting input / summing node
```

Important

Before soldering real photodiodes, verify FDS1010 anode/cathode orientation from the official drawing or by diode-test. Do not guess lead polarity from memory.

6 Transimpedance Amplifier

6.1 Why a TIA is needed

The photodiode produces current, but the ADC measures voltage. A transimpedance amplifier (TIA) converts photocurrent to voltage:

$$V_{\text{out}} \approx V_{\text{ref}} - I_{\text{PD}} R_f. \quad (16)$$

Depending on photodiode orientation and reference choice, the sign may flip. The important point is that R_f sets the current-to-voltage gain.

6.2 OPA2381 choice

The OPA2381 is a dual precision transimpedance amplifier suitable for the two photodiode channels. The frozen supply choice is:

$$V_+ = 5 \text{ V}, \quad V_- = 0 \text{ V}. \quad (17)$$

No negative rail is used in v1. The ADS8353 input range is unipolar, so a single-supply TIA is simpler and matches the ADC.

6.3 Exact OPA2381 pin names

Pin	Exact pin name	POSM use
1	V+	+5 V analog supply
2	Out B	TIA B output to ADS8353 AINP_B
3	-In B	Photodiode B summing node
4	+In B	Non-inverting reference for channel B
5	V-	Analog ground in single-supply operation
6	+In A	Non-inverting reference for channel A
7	-In A	Photodiode A summing node
8	Out A	TIA A output to ADS8353 AINP_A

6.4 Feedback resistor exercise

The student should calculate the installed R_f using:

$$R_f = \frac{V_{\text{target}}}{I_{\text{PD,max}}}. \quad (18)$$

For the frozen worst case:

$$P_{\text{PD,max}} = 200 \text{ }\mu\text{W}, \quad I_{\text{PD,max}} \approx 100 \text{ }\mu\text{A}. \quad (19)$$

Therefore, to target 2 V full-scale TIA output:

$$R_f \approx \frac{2 \text{ V}}{100 \text{ }\mu\text{A}} = 20 \text{ k}\Omega. \quad (20)$$

6.5 Feedback capacitor exercise

The feedback capacitor C_f stabilizes the TIA and limits bandwidth. A crude bandwidth estimate is

$$f_c \approx \frac{1}{2\pi R_f C_f}. \quad (21)$$

For $R_f = 20\text{ k}\Omega$ and a target bandwidth near 50 kHz,

$$C_f \approx \frac{1}{2\pi(20\text{ k}\Omega)(50\text{ kHz})} \approx 160\text{ pF}. \quad (22)$$

This is only a starting point. Real stability depends on photodiode capacitance, board parasitics, op-amp behavior, and layout. The board should include practical footprints that allow R_f and C_f to be changed after testing.

7 ADC Subsystem: ADS8353

7.1 What the ADC does

The TIA outputs are analog voltages. The ADS8353 digitizes both channels at the same time so the FPGA receives matched samples from the two PBS output ports.

The frozen ADC operating assumptions are:

- ADS8353 on the mini PCB.
- 16-bit simultaneous dual-channel sampling.
- Internal 2.5 V reference.
- Single-ended 0 V to 5 V input range using the 0 to $2V_{\text{REF}}$ mode.
- Target comfortable readout rate: 100 kS/s.
- FPGA reads raw samples over SPI-like serial interface.

7.2 Exact ADS8353 pin names

Pin	Exact pin name	POSM use
1	AINP_A	ADC channel A positive input, from TIA A output
2	AINM_A	ADC channel A negative input, connect to GND in single-ended mode
3	REFIO_A	Internal reference output/input for channel A, local decoupling
4	REFGND_A	Reference ground for channel A
5	REFGND_B	Reference ground for channel B
6	REFIO_B	Internal reference output/input for channel B, local decoupling
7	AINM_B	ADC channel B negative input, connect to GND in single-ended mode
8	AINP_B	ADC channel B positive input, from TIA B output
9	DVDD	Digital I/O supply, +3.3 V
10	SDI	Serial data input from FPGA
11	CS	Active-low chip select/frame control from FPGA
12	SCLK	Serial clock from FPGA
13	SDO_A	Serial data output for channel A to FPGA
14	SDO_B	Serial data output for channel B to FPGA
15	GND	Digital ground
16	AVDD	Analog supply, +5 V

Important

The ADS8353 does not require external CONVST, BUSY, EOC, or RESET pins in this design. The required digital interface nets are SCLK, CS, SDI, SDO_A, and SDO_B.

7.3 ADC configuration notes

The initial configuration should be:

- single-ended input mode,
- input range 0 to $2V_{REF}$,
- internal reference enabled,
- 32-clock dual-SDO readout mode.

The mini PCB should place the reference capacitors close to REFIO_A/REFGND_A and REFIO_B/REFGND_B. The TIA outputs should be routed cleanly to AINP_A and AINP_B; the ADC negative inputs AINM_A and AINM_B should be grounded for single-ended mode.

8 Mini PCB and DF40 Mezzanine Interface

8.1 Board partitioning

The mini PCB carries the sensitive analog front end. It should be mounted close to the photodiodes and connect to the base PCB through a board-to-board mezzanine connector.

Mini PCB contains	FDS1010 lead pads, 5 V photodiode bias network, OPA2381 dual TIA, feedback networks, ADS8353 ADC, ADC reference decoupling, local supply filtering.
Base PCB provides	Power rails, FPGA SPI, timestamping, data buffering, and communication with host/POSMq.

8.2 Connector family

The selected connector family is Hirose DF40. A 20-position board-to-board pair is sufficient. Suggested parts:

- Base PCB plug: DF40C-20DP-0.4V(51).
- Mini PCB receptacle: DF40C(2.0)-20DS-0.4V(51) for a 2.0 mm stack-height option.

This is a board-to-board mezzanine connection, not a cable. The mini PCB should be mechanically supported by standoffs so the DF40 connector is not carrying all mechanical load.

8.3 Exact 20-pin connector pinout

Pin	Net name	Direction / purpose
1	+5VA	Base to mini; OPA2381 supply, ADS8353 AVDD, local analog rail
2	AGND	Analog ground return
3	+5VA_BIAS	Base to mini; photodiode reverse-bias supply, may be filtered from +5VA
4	AGND	Bias/analog return
5	+3V3D	Base to mini; ADS8353 DVDD and digital I/O rail
6	DGND	Digital ground return
7	SPI_SCLK	FPGA to ADS8353 SCLK
8	DGND	Ground guard for clock
9	SPI_CS_N	FPGA to ADS8353 CS
10	DGND	Ground guard
11	SPI_SDI	FPGA to ADS8353 SDI
12	DGND	Ground guard
13	SPI_SDO_A	ADS8353 SDO_A to FPGA
14	DGND	Ground guard
15	SPI_SDO_B	ADS8353 SDO_B to FPGA
16	DGND	Ground guard
17	MINI_DETECT_N	Mini-board detect strap to base logic
18	DGND	Digital ground return
19	NC_OR_GPIO	Spare v1 pin; leave NC or future GPIO
20	DGND	Digital ground / spare guard

Important

Do not route raw photodiode summing nodes through the connector. The photodiode anode nodes connect directly to the OPA2381 inverting inputs on the mini PCB. The connector only carries power, SPI, grounds, and low-risk logic.

9 FPGA and Software Responsibilities

9.1 FPGA role

The FPGA should not be responsible for the full Stokes fit in v1. Its job is reliable data acquisition:

- generate ADS8353 SPI frames,
- read SDO_A and SDO_B,
- timestamp samples or sample blocks,
- optionally average/decimate samples,
- expose raw channel data to the host,
- provide status flags for saturation, communication errors, and low-signal conditions.

9.2 Host/POSMq role

The host software or POSMq layer should perform the flexible math:

- dark subtraction,
- channel gain correction,
- sum and difference calculation,
- normalized difference calculation,
- QWP-angle association,
- Stokes-vector fitting,
- polarization-feedback decisions.

9.3 Useful computed quantities

After calibration:

$$I_A = g_A(\text{raw}_A - d_A), \quad I_B = g_B(\text{raw}_B - d_B), \quad (23)$$

where d_A, d_B are dark offsets and g_A, g_B are gain corrections. Then

$$P_{\text{rel}} = I_A + I_B, \quad (24)$$

$$D = \frac{I_A - I_B}{I_A + I_B}. \quad (25)$$

P_{rel} is a relative power signal. D is the normalized polarization projection at the current QWP angle.

10 Calibration Plan

10.1 Dark calibration

With the beam blocked, record the raw ADC values:

$$d_A = \langle \text{raw}_A \rangle_{\text{dark}}, \quad d_B = \langle \text{raw}_B \rangle_{\text{dark}}. \quad (26)$$

This removes electronic offsets, dark current, and static room-light background.

10.2 Gain matching

The two photodiode/TIA channels will not be identical. Estimate relative gains using known polarization states or by rotating the analyzer through calibration angles. Store correction factors g_A and g_B in software.

10.3 QWP zero-angle calibration

The mechanical zero angle of the QWP will not automatically match the mathematical zero in the Stokes model. Calibrate the QWP angle by measuring known input polarization states and fitting the measured $D(\theta)$ curve.

10.4 Stokes fitting

For full-Stokes operation, step the QWP through multiple angles and record $D(\theta)$ at each angle. Fit the measured curve to the calibrated rotating-retarder model. Use more than the minimum number of angles during calibration; extra data helps reveal backlash, offsets, and nonideal optics.

11 Mechanical Design

11.1 Prototype and final materials

The prototype holder can be 3D printed. The final nice version should be machined from black anodized 6061 aluminum to match the stiffness and optical-lab feel of commercial optomechanics.

11.2 Holder requirements

The mechanical holder should include:

- BP058 sampler mount or slot,
- interface for the motorized QWP assembly,
- pocket for the 12.7 mm PBS cube,
- two FDS1010 photodiode pockets or slots,
- mini-PCB mounting points,
- standoff support between mini PCB and base PCB if stacked,
- strain relief for photodiode leads,
- optional black cover or baffle to reduce ambient light.

11.3 Ambient light

Ambient light can be subtracted during dark calibration, but it is better to reduce it mechanically. The FDS1010 active area is large, so a simple black cover or internal baffle is strongly recommended.

12 What Counts as Done

A successful first handoff should include:

- optical/mechanical CAD showing BP058, QWP, PBS, two FDS1010s, mini PCB, base PCB interface, and standoffs;
- schematic for FDS1010 bias, OPA2381 dual TIA, ADS8353, reference decoupling, power filtering, and DF40 interface;
- completed transimpedance and bandwidth calculations using the frozen power budget;
- FPGA ADS8353 SPI readout module or clear interface spec;
- raw data capture at the target 100 kS/s capability or a justified lower initial rate;
- local test points for TIA outputs and ADC references;

- dark calibration and channel gain-calibration procedure;
- demonstration with controlled input power showing ADC response scales linearly;
- demonstration with QWP/PBS showing expected sum/difference behavior;
- short report with measured noise, saturation level, and usable optical power range.

13 Suggested Repository / Project Structure

```
posm_polarization_tap/
  README.md
  docs/
    00_project_brief.md
    01_polarization_basics.md
    02_power_budget_and_tia_gain.md
    03_adc_interface.md
    04_mechanical_notes.md
    05_calibration_plan.md
    06_reading_list.md
  electronics/
    schematic/
    layout/
    datasheets/
    calculations/
  mechanical/
    cad/
    drawings/
    prints/
  fpga/
    ads8353_spi_reader.sv
    sample_timestamp.sv
    averaging_decimator.sv
    register_bank.sv
    tb_ads8353_spi_reader.sv
  software/
    capture_raw.py
    dark_calibration.py
    gain_calibration.py
    stokes_fit.py
    qwp_scan.py
  outputs/
    plots/
    calibration/
    test_data/
```

14 Resources and Reading Map

14.1 Project parts and datasheets

Topic	Resource	Why it matters
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BP058 sampler	Thorlabs BP058 product page: https://www.thorlabs.com/item/BP058	Confirms beam-sampler geometry and 8:92 split assumption.
FDS1010 photodiode	Thorlabs FDS1010 product page: https://www.thorlabs.com/item/FDS1010	Photodiode dimensions, wavelength range, and support documents.
FDS1010 specs	FDS1010 datasheet mirror: https://www.ic72.com/pdf_file/f/637186.pdf	Useful published values for capacitance, dark current, bias, and terminal behavior.
QWP	Thorlabs WPQ05M-780 product page: https://www.thorlabs.com/item/WPQ05M-780	Mechanical and optical information for the mounted 780 nm quarter-wave plate.
PBS	Shalom EO 20707-002 product page: https://www.shalomeo.com/Laser-Crystals-and-Components/Beam-Splitters/Polarizing-Cube-Beamsplitters/ZF-Glass-650-950nm-12.7x12.7x12.7mm-PBS-cube	Confirms 12.7 mm cube, 650–950 nm band, and part number.
OPA2381	TI OPA2381 datasheet: https://www.ti.com/lit/gpn/OPA2381	Pin names, supply range, TIA examples, output swing, and stability context.
ADS8353	TI ADS8353-Q1 datasheet: https://www.ti.com/lit/ds/symlink/ads8353-q1.pdf	Exact pin names, SPI timing, reference, input range, and 32-clock readout mode.
DF40 connector	Hirose DF40 catalog: https://www.hirose.com/en/product/document?clcode=&documentid=en_DF40_CAT&documenttype=Catalog&lang=en&productname=&series=DF40	Board-to-board connector dimensions, stack-height options, and footprints.
DF40 plug example	Hirose/DigiKey DF40C-20DP-0.4V(51) page: https://www.digikey.com/en/products/detail/hirose-electric-co-ltd/DF40C-20DP-0-4V-51/1969479	Suggested 20-pin plug for base PCB.
DF40 receptacle example	Hirose DF40C(2.0)-20DS-0.4V(51) page: https://www.hirose.com/product/p/CL0684-4040-0-51	Suggested 20-pin receptacle for mini PCB.

14.2 Polarization and Stokes parameters

Topic	Resource	Use
Stokes parameters	RP Photonics Encyclopedia, “Stokes parameters”: https://www.rp-photonics.com/stokes_parameters.html	Friendly conceptual reference for S_0, S_1, S_2, S_3 .
Polarization optics	Hecht, <i>Optics</i> , polarization chapters	Textbook-level background on waveplates, PBS cubes, and Jones/Stokes descriptions.

Rotating-retarder polarimetry	po-	Azzam, “Photopolarimetric measurement of the Mueller matrix by Fourier analysis of a single detected signal”	Classic reference for Fourier-style rotating-retarder polarimetry.
Practical polarimetry		Goldstein, <i>Polarized Light</i>	Deeper reference for students who want the full theory.

14.3 Photodiodes and TIAs

Topic	Resource	Use
Photodiode basics	Hamamatsu photodiode technical notes: https://www.hamamatsu.com/us/en/resources/interactive-tools/technical-notes.html	Good intuition for responsivity, dark current, capacitance, and reverse bias.
TIA basics	Texas Instruments Precision Labs / op-amp transimpedance amplifier material	Practical TIA design workflow and stability ideas.
Feedback capacitor	Analog Devices/TI application notes on photodiode amplifier compensation	Helps calculate C_f and understand why large photodiodes need compensation.
Noise basics	Horowitz and Hill, <i>The Art of Electronics</i> , photodiode amplifier sections	Useful practical analog-electronics background.

14.4 ADC, SPI, and FPGA readout

Topic	Resource	Use
ADS8353 timing	TI ADS8353-Q1 datasheet serial-interface section	Exact SCLK/CS/SDI/SDO behavior.
SPI design	FPGA vendor SPI master examples and timing diagrams	Useful for writing <code>ads8353_spi_reader.sv</code> .
Verilog/SystemVerilog	ZipCPU tutorials: https://zipcpu.com/tutorial/	Good habits for testbenches, interfaces, and streaming data.
Simulation	Verilator documentation: https://verilator.org/guide/latest/	Fast linting/simulation for SPI reader and timestamp logic.
Data calibration	Python/Numpy documentation	Dark subtraction, gain correction, and least-squares Stokes fitting.

14.5 What not to overbuild first

Do not begin with a full autonomous FPGA Stokes fitter. The first useful instrument is simpler:

```
ADS8353 -> FPGA SPI reader -> timestamped raw samples -> host software
```

The host can do the math. Once the optics, TIA, ADC, and mechanical package work, more FPGA-side processing can be added if it is genuinely useful.